Subhomogeneous Operator Systems and Classification of Operator Systems Generated by Λ-Commuting Unitaries

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Matrix Convex Sets

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Matrix Extreme Points

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Subhomogeneous Operator Systems

q-Commuting Unitaries

Matrix Convex Sets

Matrix Convex Set

A **matrix convex set** over a vector space V is a collection $K = (K_n)_{n\in\mathbb{N}}$ of subsets $K_n \subset M_n(V)$, such that:

$$\sum_i \gamma_i^* v_i \gamma_i \in K_n \tag{1}$$

whenever $v_i \in K_{n_i}$ and $\gamma_i \in M_{n_i,n}$ such that $\sum_i \gamma_i^* \gamma_i = I_n$.

- The sum (1) is called a **matrix convex combination**.
- When V is a locally convex vector space, $K = (K_n)_{n\in\mathbb{N}}$ is a **compact matrix convex set** if each K_n is compact in $M_n(V)$ in the product topology.

Matrix State Space

n-positive maps

Let $\phi : \mathcal{S} \to \mathcal{R}$ be two operator systems. ϕ is called **n-positive** if $\phi_n(s_{ij}) = (\phi(s_{ij}))$ is positive in $M_n(\mathcal{R})$ whenever (s_{ij}) is positive in $M_n(\mathcal{S})$.

- ϕ is called **completely positive** if it is n-positive for all $n \in \mathbb{N}$.
- If ϕ is completely positive with $\phi(1_{\mathcal{S}}) = 1_{\mathcal{R}}$, we call it **UCP**.
- A unital n/completely positive map with an n/completely positive inverse is called an n/complete order isomorphism.

Matrix State Space

Let S be an operator system. The **matrix state space** of S is the collection $UCP(S) = (UCP(S, M_n))_{n \in \mathbb{N}}$, where $UCP(S, M_n) = \{\phi : S \to M_n | \phi \text{ is UCP}\}$.

- $UCP(S)$ is a matrix convex set over S^*, and it is compact.
- $UCP(S)$ is one of the prime examples of compact matrix convex sets, associated with an operator system.

Matrix Affine Maps

Matrix Affine Maps

A matrix affine map from K to L is a collection $\theta = (\theta_n)_{n\in\mathbb{N}}$ of mappings $\theta_n : K_n \to L_n$ such that:

$$\theta_n\left(\sum_{i=1}^{k} \gamma_i^* v_i \gamma_i\right) = \sum_{i=1}^{k} \gamma_i^* \theta_{n_i}(v_i)\gamma_i \tag{2}$$

for all $v_i \in K_{n_i}$ and $\gamma_i \in M_{n_i,n}$ for $i = 1,\ldots,k$ satisfying $\sum_{i=1}^{k} \gamma_i^* \gamma_i = I_n$.

- When V, W are both topological vector spaces, we say that θ is continuous if θ_n is continuous for all $n \in \mathbb{N}$.
- A matrix affine map θ is called a matrix affine homeomorphism if each θ_n is a homeomorphism.

Matrix Affine Maps

- Let $\phi : \mathcal{S} \to \mathcal{R}$ be UCP. Define $\phi_n^* : \mathrm{UCP}(\mathcal{R}, M_n) \to \mathrm{UCP}(\mathcal{S}, M_n)$ by:

$$\phi_n^*(\psi) = \psi \circ \phi. \tag{3}$$

  The collection $\phi^* = (\phi_n^*)_{n \in \mathbb{N}}$ is a continuous and matrix affine map from $\mathrm{UCP}(\mathcal{R})$ and $\mathrm{UCP}(\mathcal{S})$.

- If ϕ has a UCP inverse, ϕ^* is a matrix affine homeomorphism, with $(\phi^*)^{-1} = (\phi^{-1})^*$.

Proposition

(Similar to Webster-Winkler) $T = (T_n)_{n \in \mathbb{N}} : \mathrm{UCP}(\mathcal{R}) \to \mathrm{UCP}(\mathcal{S})$ be a matrix affine homeomorphism. Then, T induces a complete order isomorphism of $\mathcal{S}$ and $\mathcal{R}$.

Matrix Extreme Points

Matrix Extreme Point

Let $K = (K_n)_{n\in\mathbb{N}}$ be a matrix convex set and let $v \in K_n$ for some $n \in \mathbb{N}$. v is a matrix extreme point in K if whenever v is a proper matrix convex combination: $v = \sum_{i=1}^{k} \gamma_i^* v_i \gamma_i$, then for all i, $n_i = n$ and $v = u_i^* v_i u_i$ for some unitary $u_i \in M_n$.

- The set of matrix extreme points in K is denoted by $\partial K = (\partial K_n)_{n\in\mathbb{N}}$.
- [Webster-Winkler, '00] K is generated by ∂K. Namely:

$$K = \overline{\left\{ \sum_i \gamma_i^* v_i \gamma_i \,\middle|\, v_i \in \partial K_{n_i}, \sum_i \gamma_i^* \gamma_i = I_n \right\}}_{n\in\mathbb{N}} \tag{4}$$

Absolute Matrix Extreme Points

Absolute Matrix Extreme Point

Let $K = (K_n)_{n\in\mathbb{N}}$ be matrix convex and let $v \in K_n$ for some $n \in \mathbb{N}$. v is an **absolute matrix extreme point** in K if whenever v is a matrix convex combination: $v = \sum_{i=1}^{k} \gamma_i^* v_i \gamma_i$ such that γ_i are all non-zero, then for all i:

- if $n_i \leqslant n$, then $n_i = n$ and $v = u_i^* v_i u_i$ for some unitary $u_i \in M_n$.
- if $n_i > n$, then there exists some $w_i \in K_{n_i - n}$ such that $v_i = u_i^*(v \oplus w_i)u_i$ for some unitary $u_i \in K_{n_i}$.

- The set of absolute matrix extreme points in K is denoted by Abex(K).
- Abex(K) might not generate K. In fact, Abex(K) might be empty.

Characterization of Matrix Extreme Points

Boundary Representations

A $*$-representation $\pi : C^*(\mathcal{S}) \to B(H)$ is a **boundary representation** for $\mathcal{S}$ if it is irreducible, and it is the unique UCP extension of $\pi|_{\mathcal{S}}$.

Proposition

(Similar to Kleski '14, Evert-Helton-Klep-McCullough '18) A point $\phi \in \mathrm{UCP}(\mathcal{S}, M_n)$ is an **absolute matrix extreme point** of $\mathrm{UCP}(\mathcal{S})$ if and only if it extends to a boundary representation on $C^*(\mathcal{S})$.

Proposition

Let $K = (K_n)_{n=1}^{\infty}$ be a compact matrix convex set over a locally convex vector space V. Then $v \in \partial K_n$ is a matrix extreme point which is not absolute if and only if it is of the form $\gamma^* \tilde{v} \gamma$ for some $\tilde{v} \in \partial K_{n+1}$.

Subhomogeneous Matrix Convex Sets

Subhomogeneous Matrix Convex Set

A compact matrix convex set $K = (K_n)_{n=1}^{\infty}$ over a locally convex vector space V is said to be Subhomogeneous if there exists some $N \in \mathbb{N}$ such that $\partial K_n = \varnothing$ for all $n > N$. If N is the smallest integer satisfying these conditions, we say that K is N-Subhomogeneous.

Theorem, K '23

Let $\mathcal{A}$ be a unital C^*-algebra. The following are equivalent:

- $\mathcal{A}$ is N-Subhomogeneous as a C^*-algebra.
- UCP$(\mathcal{A})$ is N-subhomogeneous.

Subhomogeneous Matrix Convex Sets

Subhomogeneous Operator System

An operator system $\mathcal{S}$ is said to be Subhomogeneous if $UCP(\mathcal{S})$ is Subhomogeneous, and that it is N-Subhomogeneous if $UCP(\mathcal{S})$ is N-Subhomogeneous.

Proposition

Let $K = (K_n)_{n=1}^{\infty}$ be an N-Subhomogeneous compact matrix convex set over some locally convex vector space V. Then $K = co(K_N)$.

Theorem, K '23

Let $\mathcal{S}, \mathcal{R}$ be $N_{\mathcal{S}}$-subhomogeneous and $N_{\mathcal{R}}$-subhomogeneous operator systems, and set $N = \max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$. Then, $\mathcal{S}, \mathcal{R}$ are completely order isomorphic if and only if they are N-order isomorphic.

Idea of the Proof

- If ϕ is a complete order isomorphism, it is an N-order isomorphism.
- For the converse, assume ϕ is an N-order isomorphism. For all $n \leqslant N$, $\phi_n^* : \mathrm{UCP}(\mathcal{R}, M_n) \to \mathrm{UCP}(\mathcal{S}, M_n)$ given by $\phi_n^* \psi = \psi \circ \phi$ is a continuous bijection.
- This map extends to a matrix affine homeomorphism:

$$\phi^* = (\phi_n^*)_{n \in \mathbb{N}} : \mathrm{co}(\partial \mathrm{UCP}(\mathcal{R})) \to \mathrm{co}(\partial \mathrm{UCP}(\mathcal{S}))$$

  which is a matrix affine homeomorphism of $\mathrm{UCP}(\mathcal{S})$ and $\mathrm{UCP}(\mathcal{R})$, by the previous proposition.
- By the previous theorem, $\mathcal{S}, \mathcal{R}$ are completely order isomorphic.
- In fact, one can show that ϕ is the complete order isomorphism.

Finite Dimensional Operator Systems

Proposition

Let $\mathcal{S}, \mathcal{R}$ be two finite dimensional operator systems, and assume that $\mathcal{S}$ is N-subhomogeneous. Then, $\phi : \mathcal{R} \to \mathcal{S}$ is unital and N-positive if and only if it is UCP.

- We believe this is true for infinite dimensional operator systems, but our proof uses finite dimensionality.
- Let $s \in B(\mathcal{H})$ be a self-adjoint operator. $\mathcal{S}(s)$ is 1-subhomogeneous (all irreducible representations of $C^*(s)$ are 1-dimensional).Therefore, any unital and positive map from a finite dimensional operator system to $\mathcal{S}$ is completely positive.

Operator Systems Generated by Unitaries

- Let $\mathbf{s}, \mathbf{r}$ be two d-tuples of unitaries and let $\mathcal{S}, \mathcal{R}$ be the operator systems generated by $\mathbf{s}$ and $\mathbf{r}$.
- Assume $\phi : \mathcal{S} \to \mathcal{R}$ is a unital and linear map such that $\phi(s_i) = r_i$ for $i = 1, \ldots, d$.
- It is well known that if ϕ is completely positive, it extends to a $*$-homomorphism of $C^*(\mathbf{s})$ and $C^*(\mathbf{r})$.
- In particular, ϕ is a complete order isomorphism if and only if it extends to a $*$-isomorphism of $C^*(\mathbf{s}), C^*(\mathbf{r})$.

Theorem, K '23

Assume $\mathcal{S}$ and $\mathcal{R}$ are $N_{\mathcal{S}}$-suhomogeneous and $N_{\mathcal{R}}$-subhomogeneous, respectively. For $N = \max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$, ϕ is unital and N-positive if and only if it extends to a $*$-homomorphism from $C^*(\mathbf{s})$ to $C^*(\mathbf{r})$.

q-Commuting Unitaries

- Let q be a complex number of modulus 1. We say that u, v are q-commuting unitaries if u, v are both unitaries and $uv = qvu$.
  - We call q rational if it's of the form $q = e^{2\pi i \frac{k}{n}}$ for some coprime $0 \leqslant k < n$.

Theorem, K '23

Let u, v be a pair of q-commuting unitaries with a rational $q = e^{2\pi i \frac{k}{n}}$, and set $\mathcal{S} = \mathcal{S}(u, v)$. Then, $\mathrm{UCP}(\mathcal{S})$ is n-subhomogeneous, and $\mathrm{Abex}(\mathrm{UCP}(\mathcal{S}))$ is a subset of $\mathrm{UCP}(\mathcal{S}, M_n)$. Moreover, $\partial\mathrm{UCP}(\mathcal{S}, M_k) \neq \varnothing$ for $k \leqslant n$.

Theorem, K' 23

Let u, v be a pair of q-commuting unitaries with an irrational q, and set $\mathcal{S} = \mathcal{S}(u, v)$. Then, $\mathrm{UCP}(\mathcal{S})$ is not subhomogeneous. Moreover, $\mathrm{Abex}(\mathrm{UCP}(\mathcal{S})) = \varnothing$, and $\partial\mathrm{UCP}(\mathcal{S}, M_n)$ is non-empty for all $n \in \mathbb{N}$.

q-Commuting Unitaries

Theorem, K '23

Let u, v be a pair of q-commuting unitaries with a rational $q = e^{2\pi i \frac{k}{n}}$. Let u', v' be a pair of q'-commuting unitaries.

- Assume that the mapping $u' \mapsto u, v' \mapsto v$ extends to a unital n-positive map from $\mathcal{S}(u', v')$ to $\mathcal{S}(u, v)$. Then, $q' = q$ and the mapping is completely positive.
- In addition, if the mapping is an n-order isomorphism, $C^*(u, v)$ and $C^*(u', v')$ are $*$-isomorphic.

- Note that for subhomogeneous operator systems generated by q-commuting unitaries, we replaced $\max\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$ with $\min\{N_{\mathcal{S}}, N_{\mathcal{R}}\}$.

Example

- Let $\mathcal{S}$ be the operator system generated by M_{x_1}, M_{x_2}, the multiplication by coordinate functions acting on $B(L^2(\overline{\mathbb{B}}_2))$. The pair is self-adjoint and commuting.
- (Davidson, Dor-On, Shalit, Solel) UCP$(\mathcal{S})$ is 1-subhomogeneous.
- Let $\mathcal{R}$ be the 2-subhomogeneous operator system generated by F_1, F_2, the two anti-commuting unitary matrices:

$$F_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- The map $F_1 \mapsto M_{x_1}, F_2 \mapsto M_{x_2}$ extends to a 1-order isomorphism of $\mathcal{R}, \mathcal{S}$.
- The mapping does not extend to a 2-order isomorphism of $\mathcal{R}, \mathcal{S}$.
- Therefore, taking the minimal N does not work in full generality.